

Empirical Evaluation of ProtoHedge: Interpretable Deep Hedging with Real Market Data

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ABSTRACT

Deep Hedging learns sequential hedging policies directly from market states, transaction costs, and a risk-sensitive objective, but its neural decisions are difficult to audit. ProtoHedge addresses this problem by routing each hedge through a finite set of representative market states, or prototypes: the action is determined by the current state’s similarity to those prototypes and their associated hedge responses. This paper tests whether that interpretability mechanism remains viable on historical option data, where market regimes, frictions, and non-stationarity matter. We compare ProtoHedge with classical delta-style rules and a matched black-box Deep Hedging benchmark under common costs, bounds, accounting, and model-selection controls. In the European short-call panel, ProtoHedge produces the strongest mean and tail frontiers while using less position-bound occupancy on average. Prototype use is concentrated, regime-sensitive, and traceable to concrete historical episodes, supporting the claim that interpretability can be added to deep hedging without sacrificing empirical competitiveness.

CCS CONCEPTS

• **Applied computing** → **Finance**; • **Computing methodologies** → *Instance-based learning*; *Neural networks*; *Sequential decision making*.

KEYWORDS

deep hedging, ProtoHedge, interpretable machine learning, option hedging, empirical finance

ACM Reference Format:

Imon Banerjee, Sayak Chakrabarty, and Yanis Zaim. 2018. Empirical Evaluation of ProtoHedge: Interpretable Deep Hedging with Real Market Data. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym ’XX)*. ACM, New York, NY, USA, 12 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 INTRODUCTION

Deep hedging replaces a prescribed analytical hedge with a policy learned from market states, realized trading costs, and a risk-sensitive objective. This flexibility is valuable when hedging is

discrete, costly, constrained, and exposed to dynamics that depart from a stylized pricing model. Its central practical weakness is that a neural policy can be difficult to audit: strong performance alone does not explain which market states drive a hedge, whether comparable states receive comparable actions, or whether a model has learned a stable economic rule.

ProtoHedge addresses this problem by routing each decision through a finite set of representative market states, called prototypes. The original paper established that this prototype structure can retain much of the performance of an unrestricted neural hedge in Black–Scholes and stochastic-volatility simulations. The unresolved question is empirical. Historical options data contain non-stationarity, imperfect liquidity, changing volatility, and discontinuous episodes that cannot be fully captured by a controlled simulator.

This paper evaluates ProtoHedge on historical U.S. option-hedging episodes. Our contribution is empirical rather than architectural: we preserve the fixed-prototype design and test whether its performance–interpretability tradeoff survives on real market data. The paper makes four contributions:

- (1) We construct a ten-ticker panel of 30,797 contiguous 20-step historical option-hedging episodes from WRDS-sourced underlying and option data.
- (2) We evaluate ProtoHedge against unhedged exposure, transparent delta-style rules, and a matched black-box Deep Hedging benchmark under transaction costs, hard bounds, step-wise accounting, and multi-seed validation-based selection.
- (3) We show that ProtoHedge is competitive across the panel while reducing average reliance on hard position bounds.
- (4) We test interpretability directly through prototype concentration, regime-dependent usage, and retrieval of nearest historical episodes.

2 PROTOHEDGE AND THE EMPIRICAL VALIDATION PROBLEM

2.1 Deep Hedging as Sequential Risk Management

Deep hedging replaces a prescribed replication formula with a policy learned from observed market states and a risk-sensitive hedging objective. Let $P_t \in \mathbb{R}^m$ be the vector of hedge-instrument prices, let h_t be the cumulative hedge inventory, and let τ_t denote time remaining. At time t , a policy observes a state such as

$$x_t = (P_t, h_{t-1}, \tau_t)$$

and outputs a trade increment

$$a_t = \pi_\theta(x_t), \quad h_t = \Pi_{\mathcal{H}}(h_{t-1} + a_t),$$

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Conference acronym ’XX, June 03–05, 2018, Woodstock, NY

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ACM ISBN 978-1-4503-XXXX-X/2018/06

<https://doi.org/XXXXXXX.XXXXXXX>

where $\Pi_{\mathcal{H}}$ enforces the admissible position set. For an option payoff $H(P_{0:T})$ and per-unit transaction-cost vector κ_t , the terminal gain can be written schematically as

$$G^{\pi_\theta} = -H(P_{0:T}) + \sum_{t=0}^{T-1} h_t^\top (P_{t+1} - P_t) - \sum_{t=0}^{T-1} \kappa_t^\top |a_t|.$$

The agent is trained to optimize a risk criterion, for example

$$\min_{\theta} \text{CVaR}_{\alpha}(-G^{\pi_\theta}),$$

rather than to reproduce a model-implied delta exactly. This formulation accommodates discrete rebalancing, costs, hard bounds, and dynamics that need not follow a stylized pricing model.

A standard, or black-box, Deep Hedging policy represents π_θ by a feed-forward neural network. It is a necessary performance benchmark in this paper because it has the same information, objective, costs, and constraints as ProtoHedge. Its limitation is diagnostic rather than computational: an action can be evaluated without revealing which market states motivated it or whether the policy has learned an economically stable rule.

2.2 ProtoHedge: Fixed Prototypes as an Interpretable Policy

ProtoHedge retains the deep-hedging objective but restricts action generation to a finite collection of representative states

$$\mathcal{P} = \{p_1, \dots, p_K\}.$$

In the fixed-prototype design studied here, each p_k is an observed standardized state selected from the empirical state distribution. For a standardized current state $\tilde{x}_t$, nonnegative feature weights γ , and a shared learned projection f_ϕ , ProtoHedge forms

$$z_t = f_\phi(\gamma \odot \tilde{x}_t), \quad z_k^p = f_\phi(\gamma \odot p_k),$$

then measures squared distances and converts them into normalized prototype weights:

$$d_{t,k} = \|z_t - z_k^p\|_2^2, \quad \alpha_{t,k} = \frac{\exp(-d_{t,k})}{\sum_{j=1}^K \exp(-d_{t,j})}.$$

Each prototype has an associated trainable, bounded hedge response $b_k \in \mathbb{R}^m$. The action is the similarity-weighted combination

$$a_t = \sum_{k=1}^K \alpha_{t,k} b_k.$$

This decomposition is the central ProtoHedge mechanism. For every hedge decision, the model exposes the prototypes with the largest weights, their representative market states, and their associated actions. A reviewer can therefore trace a decision to examples of historically observed states rather than only to opaque network parameters. Concentrated use of a small subset of prototypes, systematic shifts in $\alpha_{t,k}$ across regimes, and retrieval of the nearest historical episodes provide direct, testable forms of interpretability.

The anchors p_k remain fixed during training in the present study; the projection f_ϕ and the prototype actions b_k are learned. This preserves the original ProtoHedge premise that a prototype should remain tied to an actual representative state, rather than drifting to an uninterpretable parameter value. Section 3 specifies how the

prototypes are extracted, how K and similarity weighting are varied, and how the policy is trained.

2.3 Why the Empirical Test Matters

The original ProtoHedge study showed that prototype-based policies can remain competitive in Black–Scholes and stochastic-volatility simulations. Historical option data pose a materially different test: volatility and liquidity vary over time, option observations are noisy, trading is costly and constrained, and market paths contain regime shifts and discontinuities. The interpretability claim is also stronger on real data, because a useful prototype should correspond to a recognizable historical state rather than only to a simulator-generated configuration.

Accordingly, this paper evaluates whether ProtoHedge remains competitive with matched black-box Deep Hedging and transparent delta-style baselines while avoiding excessive reliance on hard position bounds. It also tests whether prototype use is concentrated, regime-dependent, and retrievable as concrete historical episodes. We conduct this evaluation for both terminal European short calls and path-dependent Asian short calls, allowing the performance–interpretability tradeoff to be assessed under two economically distinct liabilities.

3 EXPERIMENTAL FRAMEWORK

This section describes the empirical design used to evaluate ProtoHedge on historical market data. Our implementation is deliberately conservative. We use fixed-length out-of-sample hedging episodes, explicit transaction costs, bounded trading actions, bounded cumulative hedge positions, validation-based model selection, and multi-seed evaluation. The goal is not merely to train a profitable hedge in sample, but to assess whether ProtoHedge remains competitive with vanilla Deep Hedging under a realistic and reproducible real-data protocol.

3.1 Data and Episode Construction

Our real-data experiments are based on historical U.S. equity and ETF option data obtained through WRDS, combining underlying spot-price histories with daily listed option records. The final panel contains ten liquid underlyings spanning broad equity indices, sector ETFs, fixed income, and single-name equities:

$$\{\text{AAPL, IWM, NVDA, QQQ, SPY, TLT, XLE, XLF, XLK, XLV}\}.$$

For each ticker, the cleaned feature panel runs from 2005-01-11 to 2024-12-31.

The raw option-chain data are filtered to call options and aligned with the corresponding daily spot series. For each date, we select a single liquid near-at-the-money call contract by sorting candidate options according to:

- (1) absolute distance between strike and spot,
- (2) absolute distance between the contract's days-to-expiration and a 30-day target maturity,
- (3) bid–ask spread,
- (4) open interest (descending), and
- (5) volume (descending),

subject to a maturity filter of 20 to 40 calendar days to expiration. This yields a daily panel of one representative short-dated near-ATM call per underlying.

The resulting per-date feature vector is

$$x_t^{\text{raw}} = (S_t, C_t, \Delta_t^{\text{call}}, v_t, \sigma_t^{\text{impl}}),$$

where S_t is the underlying spot, C_t is the selected call price, Δ_t^{call} is the selected call delta, v_t is the selected call vega, and σ_t^{impl} is implied volatility. These features are stored in the episode tensor in the fixed order

$$(\text{spot}, \text{call_price}, \text{call_delta}, \text{call_vega}, \text{ivol}).$$

Historical hedging episodes are constructed as rolling 20-step windows. Importantly, windows are created only within contiguous trading-date blocks: whenever the gap between consecutive observation dates exceeds four calendar days, a new block is started. This prevents the episode builder from stitching together disjoint option cycles separated by large holes in the cleaned panel. Letting $\tau = 20$ denote the episode length, each final sample is a tensor in

$$\mathbb{R}^{\tau \times 5}.$$

Across the ten-ticker panel, this procedure yields 30,797 usable 20-step episodes in total.

Although the raw episode tensor contains five market features, the learning models do not consume all five directly. The hedging agents are trained on the state representation used by the deep hedging gym, namely current hedgeable prices, current hedge inventory, and time remaining. Concretely, the learned policies use the feature list

$$(\text{price}, \text{delta}, \text{time_left}),$$

where price is the two-dimensional vector of current spot and current selected call price, delta denotes the current cumulative hedge position in the two traded instruments, and time_left is the remaining time to maturity. The richer market panel remains important because it defines the liability, the classical baselines, the transaction-cost model, and the regime-analysis variables.

Finally, for numerical stability and cross-path comparability, the real-data world normalizes spot, call price, and call vega by each path's initial spot level. Under this normalization, a liability struck at-the-money at inception has normalized strike one, while remaining equivalent to a strike of S_0 in raw price units.

3.2 Hedging Tasks

We study two liabilities on the same historical episode panel.

European ATM short call. The primary task is to hedge a short at-the-money European call over a 20-step historical episode. If $S_{i,0}$ denotes the initial spot on episode i , then the strike is set to

$$K_i = S_{i,0}.$$

The liability payoff at maturity is

$$H_i^{\text{Euro}} = -\max(S_{i,T} - K_i, 0),$$

where the negative sign reflects the fact that the hedger is short the option.

Asian ATM short call. The path-dependent extension replaces the European liability with an arithmetic-average Asian call, again struck at-the-money at episode start:

$$K_i = S_{i,0}.$$

Let the episode contain T trading decisions. The Asian liability is

$$H_i^{\text{Asian}} = -\max\left(\frac{1}{T} \sum_{t=1}^T S_{i,t} - K_i, 0\right).$$

In the implementation used for the final paper sweep, the averaging window spans the full historical episode.

Hedge instruments and actions. The hedge may trade in two liquid instruments:

- (1) the underlying spot instrument, and
- (2) the selected near-ATM listed call option.

The action at each step is therefore two-dimensional,

$$a_t = (a_t^{(S)}, a_t^{(C)}),$$

where each component denotes a trade increment rather than an absolute target position. The cumulative hedge inventory evolves recursively as

$$\delta_t = \delta_{t-1} + a_t.$$

Step-wise hedge accounting. All reported experiments use step-wise hedge accounting rather than a terminal shortcut. If ΔS_t and ΔC_t denote the next-step changes in the spot and option hedge instruments, then the pathwise hedge P&L is accumulated as

$$\text{PnL}_i = \sum_{t=1}^T (a_{i,t}^{(S)} \Delta S_{i,t} + a_{i,t}^{(C)} \Delta C_{i,t}).$$

This choice is important empirically. In earlier exploratory real-data runs, more permissive accounting and looser constraints could produce apparently strong results together with unrealistic cumulative-bound saturation. The current step-wise setup is intentionally stricter and better aligned with the actual sequence of hedge rebalancing decisions.

Transaction costs. Trading costs are imposed separately on spot and option trades. The per-unit spot trading-cost coefficient is

$$c_{i,t}^{(S)} = 0.0002 |S_{i,t}|.$$

The per-unit option trading-cost coefficient is

$$c_{i,t}^{(C)} = 0.02 |v_{i,t}| + 0.0002 |\Delta_{i,t}^{\text{call}}| + 0.0005 |C_{i,t}|.$$

The realized trading cost on episode i is therefore

$$\text{Cost}_i = \sum_{t=1}^T (|a_{i,t}^{(S)}| c_{i,t}^{(S)} + |a_{i,t}^{(C)}| c_{i,t}^{(C)}).$$

Gains. For both liabilities, the terminal hedging gain is defined as

$$G_i = H_i + \text{PnL}_i - \text{Cost}_i.$$

This is the central outcome used throughout the empirical evaluation.

3.3 Models and Baselines

We compare ProtoHedge against both classical baselines and a black-box neural benchmark.

Unhedged. This baseline takes no hedge action:

$$a_t \equiv 0.$$

Its gain is simply the liability payoff.

Spot-Delta. This is a rule-based baseline that hedges only with the underlying spot instrument. At each step, the target cumulative spot position is set equal to the observed delta of the selected near-ATM call. The option hedge position remains zero. This baseline uses the market-provided option delta directly and therefore provides a transparent classical reference point.

Spot-Delta Band. This baseline extends Spot-Delta with a no-trade band. Instead of rebalancing to the point target at every step, the current cumulative spot position is left unchanged as long as it remains within

$$[\Delta_t^{\text{call}} - b, \Delta_t^{\text{call}} + b],$$

where $b \geq 0$ is a band width. If the current position exits the band, it is projected back to the closest boundary. We tune b on the validation set using the grid

$$b \in \{0.00, 0.01, 0.02, 0.05, 0.10, 0.15\},$$

selecting the band that maximizes validation mean gains.

Vanilla Deep Hedging. The main black-box benchmark is a feed-forward neural hedging policy trained end-to-end under the same world, friction model, and objective as ProtoHedge. The vanilla agent uses the state representation

$$(\text{price}, \text{delta}, \text{time_left}),$$

and outputs a two-dimensional trade increment. In the reported panel runs, the vanilla network has width 20, depth 3, and Softplus activations.

ProtoHedge. ProtoHedge uses the same state representation and output space as vanilla Deep Hedging, but constrains action generation through a prototype layer. Each current state is compared to a finite collection of stored prototypes, and the resulting similarity pattern determines the hedge action. In the reported full grid, we vary:

- (1) the number of prototypes

$$K \in \{10, 25, 50, 100\},$$

- (2) the prototype source policy, using either a trained vanilla hedge or the Spot-Delta baseline as the source from which prototypes are extracted, and
- (3) whether similarity is weighted or unweighted.

Prototype extraction follows the same reusable pipeline throughout the repo. For a given source policy, we:

- (1) construct the flattened feature matrix seen by the policy,
- (2) standardize it with a StandardScaler,
- (3) apply K -means clustering in standardized feature space, and
- (4) store, for each cluster, the nearest real standardized feature vector rather than the raw cluster centroid.

This last step matters because it keeps each stored prototype tied to an actually observed market state rather than to an abstract average point with no direct empirical counterpart.

In the reported real-data panel runs, learned distance-feature weights are held fixed, i.e. we do not yet let prototypes or feature-weight vectors move during training. This keeps the present paper focused on empirical validation of the original fixed-prototype design.

3.4 Training and Model Selection Protocol

All real-data neural models are trained under the same multi-seed protocol.

Train/validation/test splits. For each ticker and each random seed, the episode tensor is randomly permuted at the path level and split into

$$70\% \text{ train}, \quad 15\% \text{ validation}, \quad 15\% \text{ test}.$$

The main reported panel uses the three seeds

$$(1234, 2345, 3456).$$

Optimization. All neural models are trained for 800 epochs using the same CVaR-based deep hedging objective with risk-aversion parameter one. The optimizer uses learning rate

$$10^{-3},$$

together with clipping safeguards already built into the training pipeline. Classical baselines are not trained in the neural sense, but they are evaluated on exactly the same splits and under the same friction model.

Feature normalization and state semantics. The real-data world is run in normalized mode. Spot, call price, and call vega are divided pathwise by the initial spot, while the state passed to the learning agent at each step consists of current hedgeable prices, current hedge inventory, and time remaining. This keeps the neural state compact and comparable across underlyings while preserving the original market information needed for payoff definition, baseline construction, and regime analysis.

Validation-based checkpoint selection. Model selection is based on the validation split, not on the training loss alone. Let L_{val} denote the validation objective. For the final robust panel runs, checkpoints are ranked by the selection score

$$S = L_{\text{val}} + \alpha_a \overline{|a|} + \alpha_\delta \overline{|\delta|} + \alpha_b \text{BO} + \alpha_p \text{PT},$$

where $\overline{|a|}$ is mean absolute action magnitude on the validation set, $\overline{|\delta|}$ is mean absolute cumulative position magnitude, BO is validation bound occupancy, and PT is the fraction of validation paths that touch any cumulative bound. In the reported full runs, we use

$$\alpha_a = 0.005, \quad \alpha_\delta = 0.010, \quad \alpha_b = 0.100, \quad \alpha_p = 0.$$

Thus, validation selection explicitly favors models that do not rely excessively on large actions, large cumulative positions, or persistent boundary saturation.

Regularization. In addition to validation-time selection penalties, the training loss includes mild regularization terms on action and cumulative-position magnitude:

$$\lambda_a = 0.001, \quad \lambda_\delta = 0.002.$$

Concretely, if L_{hedge} is the base objective, then the optimized loss is

$$L_{\text{train}} = L_{\text{hedge}} + \lambda_a \mathbb{E}[|a|] + \lambda_\delta \mathbb{E}[|\delta|].$$

These penalties are deliberately mild: they do not dominate the objective, but they discourage the model from achieving performance through unnecessarily aggressive trading.

Trade bounds and cumulative position bounds. The final reported real-data runs impose both action-level and cumulative-position bounds. At every step and for each traded instrument, trade increments are clipped to

$$[-1, 1],$$

and cumulative hedge positions are also constrained to

$$[-1, 1].$$

These bounds are scientifically important. In exploratory unconstrained runs, apparent performance improvements could be accompanied by heavy saturation at the position boundaries. The bounded protocol is designed to test whether ProtoHedge remains competitive without relying on unrealistic edge-of-feasible-region behavior.

Paper sweep design. For the European-call panel, the final full sweep evaluates the full ProtoHedge grid described above across all ten tickers and all three seeds. The Asian-call extension uses the same empirical framework and, in the final notebook implementation, can be run with the same grid and the same seed schedule. This allows direct comparison between terminal and path-dependent liabilities without changing the training pipeline.

3.5 Evaluation Metrics

We evaluate models at both the hedging-performance level and the interpretability level.

Primary performance metrics. For each fitted model and each test split, we compute:

- (1) mean gains,

$$\mathbb{E}[G],$$

- (2) the empirical 5% Value-at-Risk and Conditional Value-at-Risk of gains,

$$\text{VaR}_{5\%}(G), \quad \text{CVaR}_{5\%}(G),$$

- (3) shortfall probability,

$$\mathbb{P}(G < 0),$$

and

- (4) realized mean trading-cost and hedge-P&L summaries.

In the paper narrative, mean gains and $\text{CVaR}_{5\%}$ are the headline performance measures. Shortfall probability is retained as a secondary summary rather than the sole criterion of success.

Position-bound diagnostics. Because constrained hedging behavior is a central part of the empirical question, we also record:

- (1) the fraction of all path-step-instrument positions lying at either cumulative position bound,
- (2) the fraction lying specifically at the upper or lower bound, and
- (3) the fraction of paths that touch any cumulative position bound at least once.

We refer to the first quantity as *bound occupancy*. Lower bound occupancy is interpreted as evidence that a model is operating more consistently in the interior of the feasible region, which is economically preferable all else equal.

Robust-screen criterion. When constructing paper frontiers among ProtoHedge models, we apply a robustness screen based on average bound occupancy and path-touch behavior. In the final reported panel setup, models with excessive bound saturation are excluded from the empirical frontier even if they perform well on raw mean gains. This prevents the final comparison from being driven by unrealistic edge-seeking solutions.

Prototype concentration metrics. For ProtoHedge models, we further compute:

- (1) average prototype weight across all paths and time steps,
- (2) top-1 prototype frequency, i.e. the proportion of decisions for which each prototype receives the highest weight, and
- (3) cumulative usage mass of the top- k prototypes.

These metrics quantify whether decisions are routed through a small number of dominant prototype states or diffusely across many prototypes.

Regime-sensitive interpretability metrics. To test whether prototype use changes systematically with market conditions, we form episode-level regime labels from the historical paths:

- *return regime*, using terciles of terminal return,
- *volatility regime*, using terciles of average implied volatility, and
- *drawdown regime*, using terciles of maximum realized drawdown.

This yields labels of the form

$$\{\text{down, middle, up}\}, \quad \{\text{low_vol, mid_vol, high_vol}\}, \quad \{\text{mild_dd, mid_dd, deep_dd}\}$$

Within each regime, we summarize gains, tail risk, shortfall, and average prototype weights. This lets us test whether the same few prototypes dominate everywhere, or whether ProtoHedge reallocates attention across market states in a regime-sensitive way.

Nearest historical states. Finally, for the dominant prototypes, we recover the nearest historical feature vectors and corresponding underlying paths. This allows us to move beyond abstract prototype coordinates and ask whether the most-used prototypes correspond to recognizable empirical market scenarios. This step is central to the interpretability claim of the paper: the prototype mechanism is not merely sparse, but empirically inspectable.

4 REAL-DATA EMPIRICAL RESULTS

This section evaluates whether ProtoHedge preserves a favorable performance-interpretability tradeoff on historical option-hedging

episodes. All reported European-option results use the common protocol described in Section 3: three random seeds, held-out test episodes, step-wise hedge accounting, transaction costs, hard trade and cumulative-position bounds, and validation-based checkpoint selection. For each ticker, results are first averaged over seeds and then aggregated with equal weight across the ten tickers.

4.1 European-Option Panel Results

Table 1 reports the main panel-level comparison. The mean-oriented ProtoHedge frontier achieves the best average mean gains across the ten-ticker panel, while the tail-oriented ProtoHedge frontier achieves the best average empirical CVaR_{5%}. The result is therefore not that one ProtoHedge configuration dominates every criterion. Rather, the prototype model produces an interpretable mean–tail frontier that is at least as competitive as vanilla Deep Hedging under both objectives.

All models face the same short-call liability, transaction-cost model, and position constraints. Thus, the relative comparison is the central object of interest. The negative level of average gains reflects the conservative constrained evaluation environment and should not be interpreted as evidence that the ordering among hedges is uninformative. Under this common protocol, a less negative gain or CVaR value is better.

Relative to vanilla Deep Hedging, the mean-oriented ProtoHedge frontier has a positive average mean-gain difference of 0.000625 and a positive average CVaR_{5%} difference of 0.001030. The tail-oriented frontier has a positive average mean-gain difference of 0.000132 and a larger average CVaR_{5%} improvement of 0.002672. The mean-oriented frontier exceeds vanilla on mean gains in all ten tickers and on CVaR_{5%} in seven tickers. The tail-oriented frontier exceeds vanilla on mean gains in six tickers and improves CVaR_{5%} in all ten tickers.

Table 2 gives the ticker-level comparison underlying these aggregate statements. The mean frontier is reported beside vanilla for average gains, while the tail frontier is reported beside vanilla for CVaR_{5%} and bound occupancy. This presentation makes the intended tradeoff explicit: ProtoHedge is not being evaluated only on a pooled average, but across broad-market, sector, fixed-income, and single-stock underlyings.

Figure 1 visualizes the mean-gain results by ticker and model. Each row should be read horizontally: the cell values are seed-averaged held-out gains for one ticker, and less negative values indicate better performance. The figure shows that the mean-oriented ProtoHedge frontier tracks or improves on vanilla Deep Hedging across the panel, with the clearest gains on AAPL, NVDA, TLT, and XLK.

Figure 2 gives the corresponding tail-risk comparison. It shows that a ProtoHedge frontier is the strongest CVaR_{5%} model for every ticker in the panel. In particular, the tail-oriented configuration improves on vanilla in all ten direct comparisons, which is the strongest panel-level evidence that prototype structure need not come at the cost of tail behavior.

4.2 Position-Bound Analysis

Position bounds prevent a learned hedge from taking unrealistically large cumulative exposures. Bound occupancy therefore measures

how often a policy is effectively pressed against the edge of the feasible trading region. A policy with strong performance but persistent bound-saturation is less economically credible, because its apparent performance may depend heavily on the imposed constraint rather than on a stable hedge rule.

The aggregate evidence favors ProtoHedge. Vanilla Deep Hedging has mean bound occupancy of 0.0775, compared with 0.0486 for the mean-oriented ProtoHedge frontier and 0.0304 for the tail-oriented frontier. These correspond to average reductions of approximately 37% and 61%, respectively, relative to vanilla. The improvement is not universal: for TLT, XLE, XLF, and XLV, the selected tail frontier has higher occupancy than vanilla. This heterogeneity is important and shows why the paper reports per-ticker diagnostics rather than only a pooled average.

Figure 3 makes this diagnostic transparent. The learned-model columns should be compared within each row; darker or larger values correspond to more frequent contact with a hard cumulative-position limit. ProtoHedge often achieves competitive mean or tail performance without greater bound reliance, particularly for AAPL, IWM, NVDA, QQQ, and SPY.

The bound analysis strengthens the main performance result. ProtoHedge does not obtain its favorable panel-level mean and CVaR performance by systematically taking more extreme feasible positions. Instead, the prototype bottleneck is consistent with a less boundary-driven policy on average, while preserving competitive hedging outcomes.

4.3 Interpretability Verification

Performance alone does not establish that ProtoHedge is interpretable. We therefore inspect the learned prototype mechanism directly using a representative QQQ frontier model with 25 prototypes. The goal is not to claim that a small number of states completely explains every decision, but to test whether the policy actually routes decisions through a sparse and inspectable set of recurring empirical states.

Figure 4 reports cumulative prototype usage after sorting prototypes from most to least used. The three most-used prototypes account for 58.4% of total assignment mass, and the ten most-used prototypes account for approximately 99.6%. Thus, although the model is supplied with 25 prototypes, its decisions are effectively concentrated in a much smaller subset of recurrently used state representatives.

Figure 5 tests whether this usage is state-dependent rather than static. Each row is a volatility regime and each column is a prototype; color denotes the average similarity weight assigned in that regime. Several dominant prototypes change substantially across regimes. For example, prototype 8 is most prominent in low-volatility episodes, whereas prototypes 18 and 23 receive substantially more weight in high-volatility episodes. This behavior is consistent with the intended interpretation of ProtoHedge: distinct representative states become relevant under different market conditions.

Finally, Figure 6 grounds the dominant prototypes in actual historical episodes. Each panel shows the normalized spot path nearest to one of the most-used prototypes, and the dashed vertical line marks the time step at which the prototype match occurs.

Model	Mean gains	CVaR _{5%}	Shortfall probability	Bound occupancy
Unhedged	-0.030502	-0.140291	0.615941	0.000000
Spot-Delta	-0.030331	-0.137235	0.815792	0.000000
Spot-Delta Band	-0.030295	-0.137247	0.815806	0.000000
Vanilla Deep Hedging	-0.030277	-0.134793	0.979211	0.077515
ProtoHedge (Mean frontier)	-0.029651	-0.133763	0.896937	0.048616
ProtoHedge (Tail frontier)	-0.030145	-0.132121	0.918602	0.030434

Table 1: Average European-option results across the ten-ticker panel. Higher mean gains and higher CVaR_{5%} values are better; lower bound occupancy is better. Shortfall probability is reported as a secondary distributional diagnostic and should be interpreted jointly with the mean and tail-loss measures.

Ticker	Mean gains		CVaR _{5%}		Bound occupancy	
	Vanilla	Proto Mean	Vanilla	Proto Tail	Vanilla	Proto Tail
AAPL	-0.03963	-0.03787	-0.14793	-0.14637	0.12504	0.00959
IWM	-0.02711	-0.02699	-0.12398	-0.12251	0.09070	0.00275
NVDA	-0.07238	-0.06988	-0.32653	-0.32036	0.17440	0.01160
QQQ	-0.02881	-0.02863	-0.11040	-0.10687	0.16596	0.03044
SPY	-0.02168	-0.02162	-0.08590	-0.08397	0.19267	0.00015
TLT	-0.01430	-0.01339	-0.08194	-0.08115	0.00000	0.08351
XLE	-0.02856	-0.02851	-0.17052	-0.16881	0.00000	0.01507
XLK	-0.02381	-0.02367	-0.10478	-0.10208	0.00000	0.13615
XLV	-0.02809	-0.02773	-0.10933	-0.10643	0.02638	0.01186
XLV	-0.01842	-0.01822	-0.08661	-0.08266	0.00000	0.00323

Table 2: Ticker-level European-option comparison with vanilla Deep Hedging. Higher mean gains and CVaR_{5%} are better, while lower bound occupancy is better. The two ProtoHedge columns use the screened mean and tail frontiers, respectively.

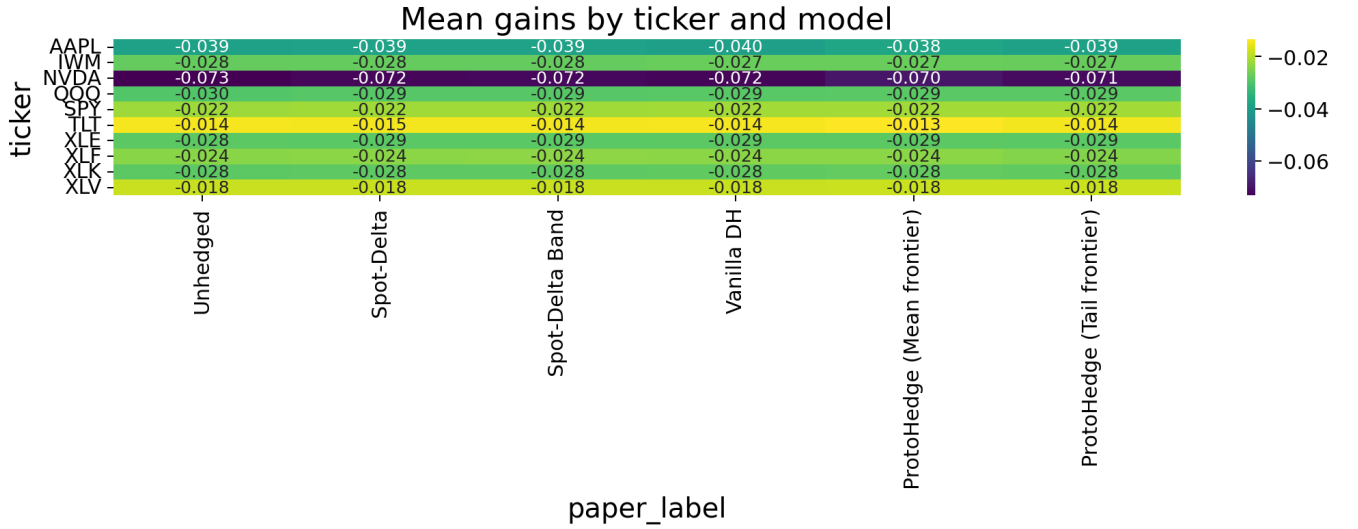


Figure 1: Mean gains by ticker and model for European short-call hedging. Each row is a ticker and each cell is the mean held-out gain averaged over seeds. Comparisons should be made horizontally within a ticker: higher, or less negative, values are better.

The retrieved examples are recognizable market configurations—including moderately rising, recovering, and sharply drawn-down paths—rather than abstract latent coordinates. This retrieval property is the practical interpretability advantage of ProtoHedge: a

hedge decision can be linked back to concrete historical situations that a domain expert can inspect.

Together, the concentration, regime-usage, and nearest-episode diagnostics provide direct evidence that the prototype layer is active

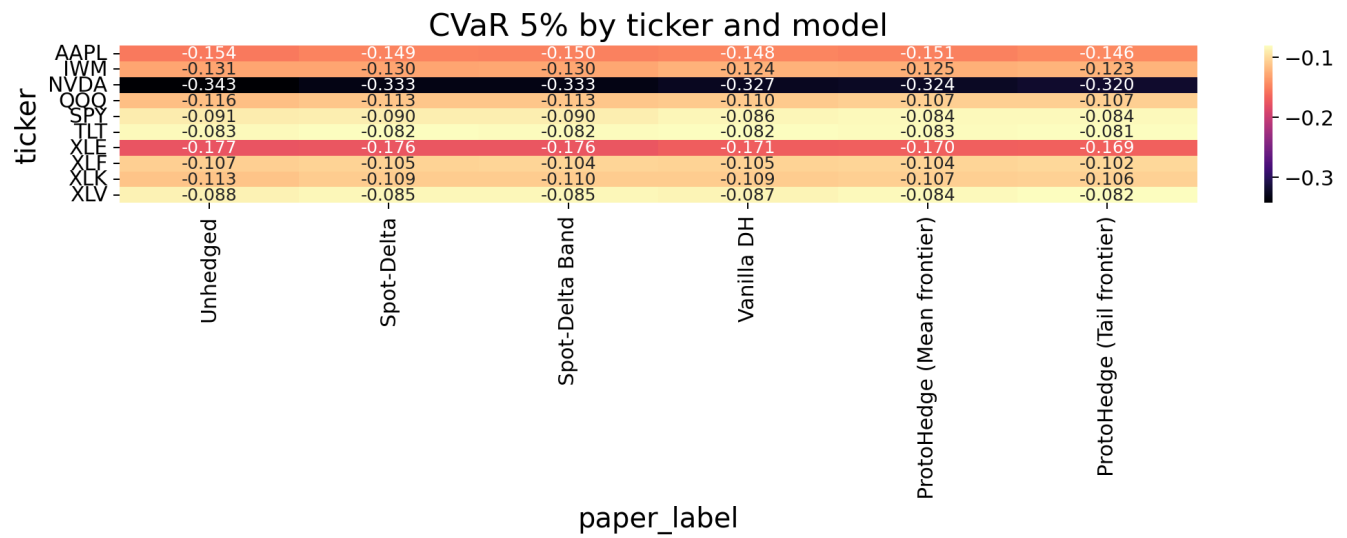


Figure 2: Empirical CVaR_{5%} by ticker and model for European short-call hedging. Less negative CVaR_{5%} values indicate smaller average losses in the worst five percent of held-out episodes. A ProtoHedge frontier provides the strongest CVaR_{5%} result for every ticker.

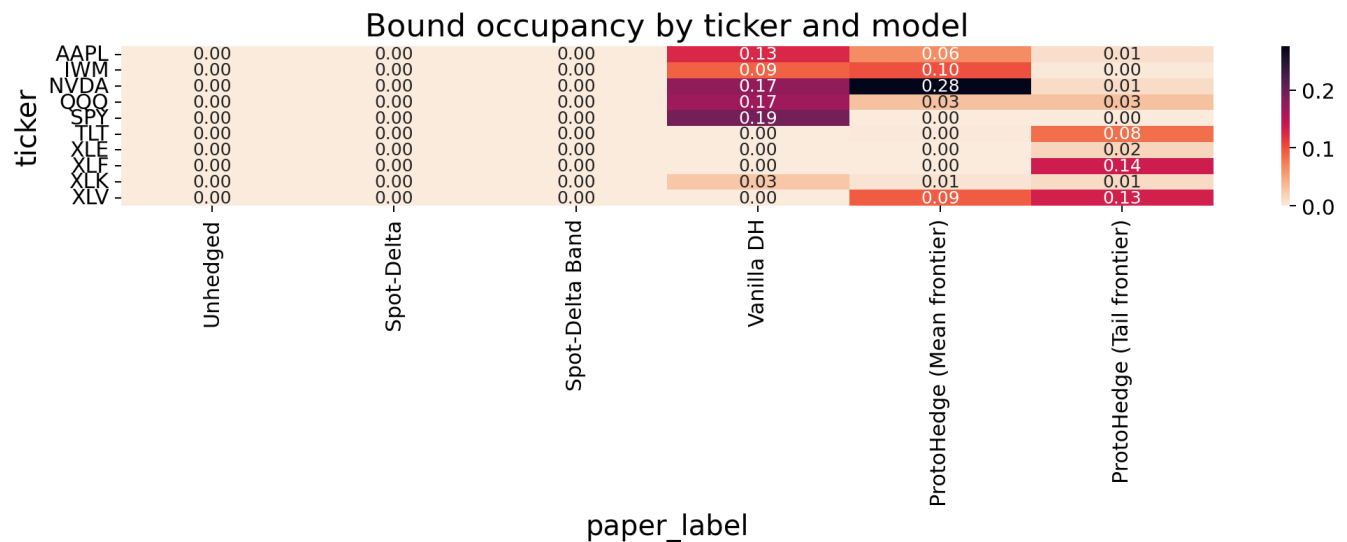


Figure 3: Cumulative position-bound occupancy by ticker and model. Each cell is the fraction of held-out decision times at which the cumulative hedge is at a hard position limit. Lower values are more economically credible. The comparison among learned policies shows that ProtoHedge often reduces reliance on the hard bounds relative to vanilla Deep Hedging.

and inspectable in the real-data setting. They do not prove that every prototype has a unique economic interpretation, but they rule out the simpler concern that prototypes are merely decorative parameters with diffuse or state-invariant usage.

4.4 Asian-Option Results

The final empirical extension applies the same framework to an at-the-money short Asian call. Unlike the European liability, the

Asian payoff depends on the arithmetic average of the underlying over the specified averaging window. This experiment therefore tests whether ProtoHedge remains competitive when the liability depends on the path rather than solely on the terminal underlying price.

The Asian-option panel will use the same ten tickers, episode construction, train-validation-test splits, three-seed protocol, transaction costs, position constraints, model grid, and interpretability

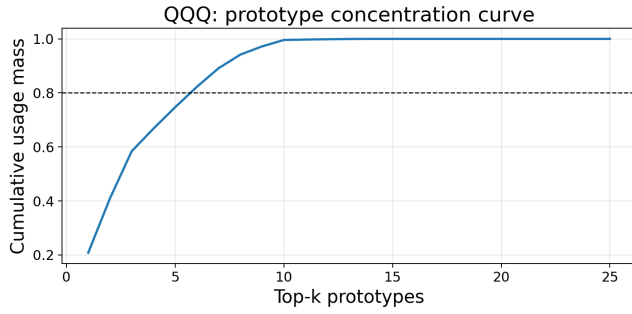


Figure 4: Prototype concentration for a representative QQQ ProtoHedge model with 25 prototypes. The horizontal axis gives the number of most-used prototypes retained, and the vertical axis gives their cumulative assignment mass. The steep initial increase shows that a small subset of prototypes explains most decisions.

diagnostics as the European experiment. The analysis will answer two questions:

- (1) Does ProtoHedge remain competitive with vanilla Deep Hedging and the delta baselines on mean gains and CVaR_{5%} when hedging a path-dependent liability?
- (2) Do the learned prototypes remain concentrated, regime-aware, and traceable to coherent historical paths when the payoff depends on path history?

5 DISCUSSION AND LIMITATIONS

The evidence in this paper supports the empirical-validation motivation of ProtoHedge. On a ten-ticker panel of historical European-option hedging episodes, ProtoHedge remains competitive with standard Deep Hedging while retaining a directly inspectable prototype layer. The mean-oriented ProtoHedge frontier achieves the strongest average mean gains across the panel, while the tail-oriented frontier achieves the strongest average empirical CVaR_{5%}. Both ProtoHedge frontiers also have lower average position-bound occupancy than standard Deep Hedging.

The appropriate interpretation is not that ProtoHedge universally dominates all hedging methods. Rather, the results show that imposing a prototype bottleneck need not cause a large empirical performance loss. In the present setting, the bottleneck can coincide with slightly better average gains or tail outcomes, while making the learned policy more auditable. This is the central performance–interpretability tradeoff that ProtoHedge was designed to deliver.

5.1 Where ProtoHedge Works

ProtoHedge performs particularly well on the criteria most closely aligned with the paper’s motivation. First, the mean-oriented frontier improves on standard Deep Hedging in average mean gains for all ten tickers. Second, the tail-oriented frontier improves empirical CVaR_{5%} relative to standard Deep Hedging for all ten tickers. These results span broad-market ETFs, sector ETFs, a Treasury ETF, and large-cap individual equities, suggesting that the finding is not tied to a single underlying.

The position-bound analysis provides a second positive result. On average, both ProtoHedge frontiers spend less time at hard cumulative position limits than standard Deep Hedging. This matters because a policy that repeatedly sits at a constraint may derive apparent performance from extreme feasible exposures rather than from a stable hedging rule. The lower average bound occupancy of ProtoHedge therefore strengthens, rather than weakens, the interpretation of its competitive performance.

Finally, the prototype diagnostics show that the interpretability mechanism remains active on real data. In the representative QQQ analysis, most prototype usage mass is concentrated in a small subset of the available prototypes, the dominant prototypes shift across return, volatility, and drawdown regimes, and prototype matches can be retrieved as concrete historical price paths. This does not mean that every prototype has a single universal economic label. It does mean that the policy can be examined through recurring market states instead of only through opaque neural-network weights.

5.2 Where ProtoHedge Does Not Dominate

The results do not support a claim of universal superiority. The relative advantage varies by ticker, metric, and selected frontier. For example, the tail-oriented ProtoHedge frontier has higher bound occupancy than standard Deep Hedging for several tickers, including TLT, XLE, XLF, and XLV. Similarly, the simple unhedged and delta-style baselines remain competitive on some secondary metrics, particularly shortfall probability under the current gain definition.

This variation is economically plausible. Hedging performance depends on the underlying’s return distribution, implied-volatility behavior, liquidity, option-market quality, and the interaction between transaction costs and position constraints. ProtoHedge should therefore be understood as an interpretable alternative with a favorable average tradeoff, not as a replacement for every black-box or classical hedge in every market.

The current results also do not establish that a single fixed ProtoHedge configuration is optimal across all assets. The reported mean and tail frontiers summarize screened configurations within each ticker. A practitioner seeking to deploy one model specification across markets would need an additional validation exercise that fixes the architecture and hyperparameters before the final test evaluation.

5.3 Limitations

Several limitations define the scope of the present contribution.

First, the study uses a historical U.S. equity and ETF option panel. Although the ten-ticker design is materially broader than a single-underlying study, it does not establish robustness across international equities, equity indices, commodities, foreign exchange, rates, or cryptocurrency markets. The results should therefore not be read as evidence that ProtoHedge is already asset-class invariant.

Second, the hedging environment is realistic in important respects—it uses historical option observations, transaction costs, step-wise accounting, and hard trading constraints—but it remains a controlled backtest. It does not fully model bid–ask execution uncertainty, market impact, funding costs, changing liquidity, margin requirements, or the operational constraints of a live trading desk.

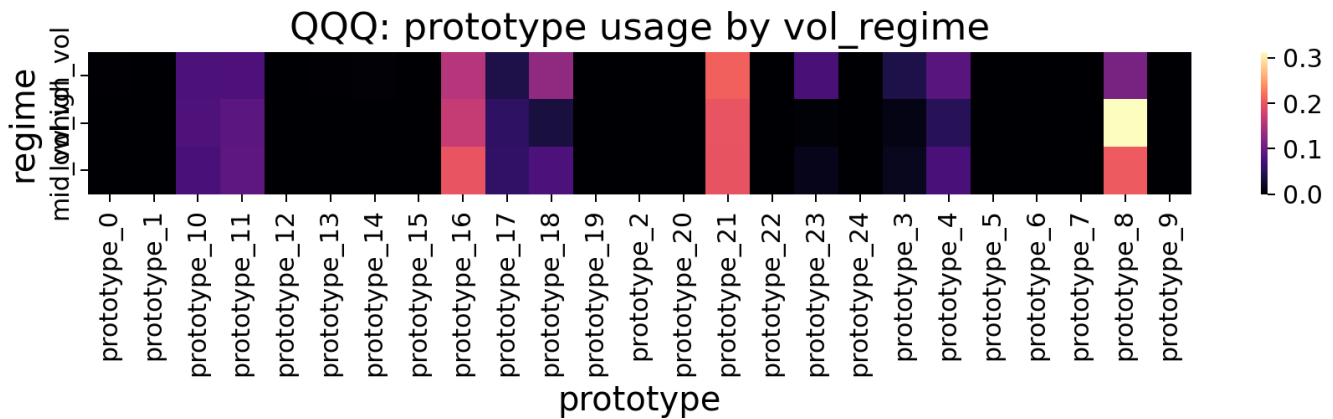


Figure 5: Prototype usage by realized-volatility regime for the representative QQQ model. Rows correspond to low-, medium-, and high-volatility held-out episodes; columns correspond to individual prototypes; color represents average prototype-assignment weight. Regime-specific changes in the dominant columns show that the prototype layer responds to market conditions rather than applying a fixed mixture of prototypes.

The short-call liability is also stylized rather than a complete portfolio of actual client positions.

Third, the current rolling 20-step episode design is well suited to controlled comparison, but future work should include strictly time-ordered out-of-sample tests with non-overlapping evaluation periods. Such tests would provide a stronger assessment of robustness to non-stationarity and would better isolate performance during specific stress events.

Fourth, the interpretability analysis is evidence of inspectability, not external validation of usability. Prototype concentration, regime-dependent usage, and nearest historical states demonstrate that the model exposes meaningful internal structure. However, this paper does not include an evaluation by professional hedgers, portfolio managers, or risk officers. Whether practitioners find the retrieved prototype states useful for actual decision support remains an open question.

Fifth, the study compares ProtoHedge with standard Deep Hedging and classical delta-style baselines, but not with the full range of recent deep-hedging architectures. In particular, future work should compare against deeper or recurrent hedging networks, alternative risk-aware objectives, and models evaluated under richer dynamics such as Chiarella–Heston-type environments. Those comparisons are necessary to locate ProtoHedge more precisely within the broader deep-hedging literature.

Finally, the present work does not provide a systematic scalability study. We evaluate several prototype counts and similarity specifications, but we do not yet fully characterize how computational cost, prototype count, feature dimensionality, and performance interact as the problem grows. This is especially relevant for high-dimensional hedging tasks involving multiple correlated assets, richer option surfaces, or longer horizons. Approximate nearest-neighbor methods, more efficient prototype-selection procedures, and trainable prototype locations are promising directions for addressing this limitation.

5.4 Path-Dependent and Practical Extensions

The final Asian-option experiment directly addresses the next empirical extension proposed in the original ProtoHedge research agenda. Because an Asian payoff depends on the trajectory of the underlying rather than only its terminal value, it provides a more demanding test of whether prototype-based decisions remain useful under path dependence. Once the full ten-ticker Asian panel is complete, it should be evaluated using the same performance, bound-occupancy, and interpretability framework as the European-option results.

Beyond Asian options, barrier options and other path-dependent liabilities would provide natural stress tests of the framework. Further empirical work should also identify whether dominant prototypes correspond to recognizable market events, including crisis periods, volatility shocks, and rapid drawdowns. The strongest eventual external validation would combine this quantitative analysis with structured feedback from professional hedgers who can assess whether prototype retrievals are economically meaningful and operationally useful.

6 CONCLUSION

Each author must be defined separately for accurate metadata identification. As an exception, multiple authors may share one affiliation. Authors' names should not be abbreviated; use full first names wherever possible. Include authors' e-mail addresses whenever possible.

ETHICS AND PRIVACY STATEMENT

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A RESEARCH METHODS

A.1 Part One

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A.2 Part Two

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B ONLINE RESOURCES

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Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009

QQQ: nearest historical episodes for top prototypes

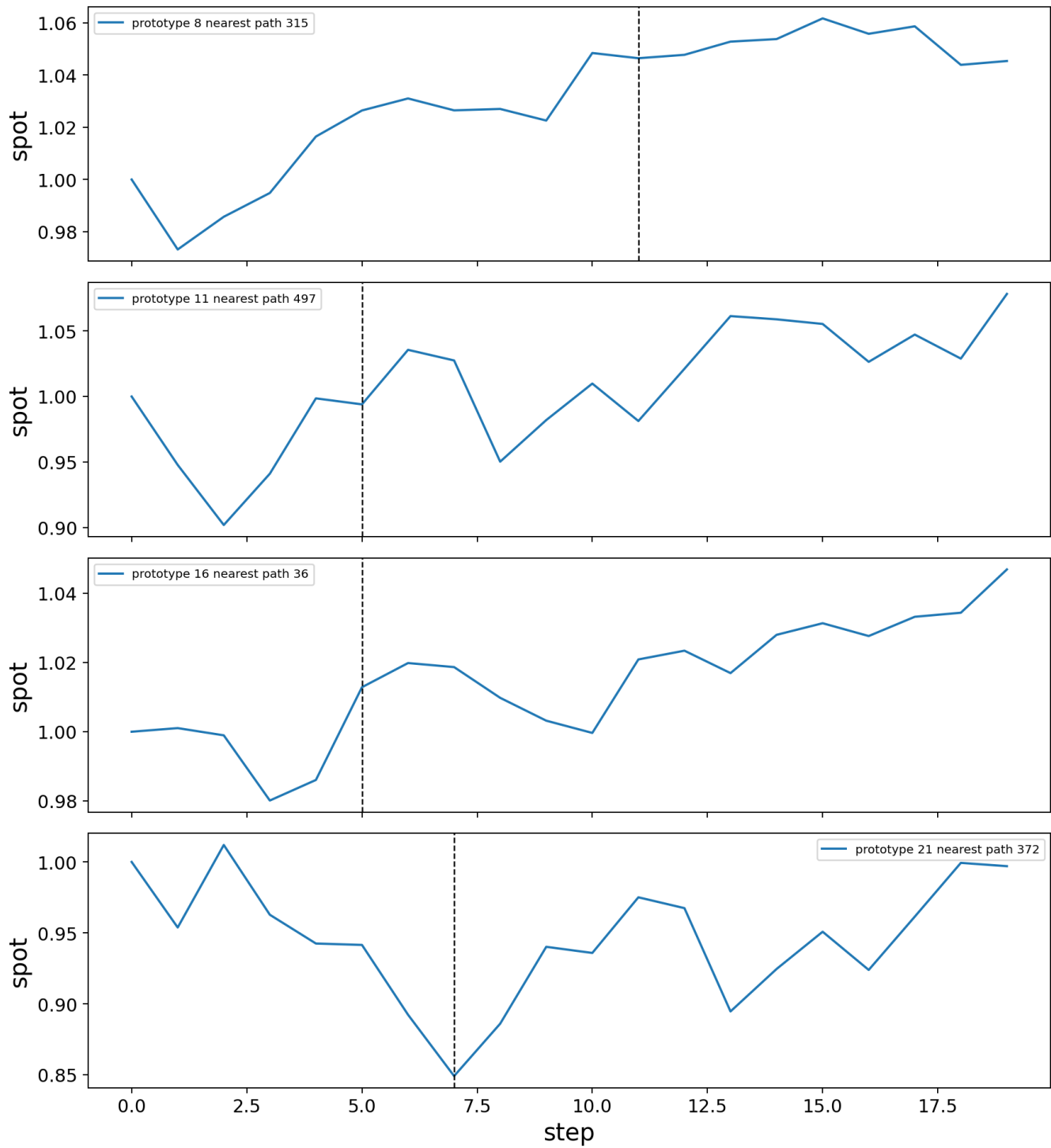


Figure 6: Nearest historical QQQ episodes for four dominant prototypes. Each subplot shows a normalized 20-step spot path; the dashed line marks the state that is nearest to the corresponding prototype. These retrievals connect the learned prototype representation to concrete historical market trajectories.